A04 · Auto Loader schema enforcement and evolution

Objective: use Auto Loader with schema enforcement and evolution.
Free Edition: Unity Catalog volume source, serverless notebook, trigger(availableNow=True). Fictional orders only.

1. schemaLocation turns on inference + evolution

```
df = (spark.readStream.format("cloudFiles")
      .option("cloudFiles.format", "csv")
      .option("cloudFiles.schemaLocation", chk)     # schema kept in chk/_schemas
      .load("/Volumes/workspace/bt_autoloader_xxxx/landing/orders"))
```

Inference samples the first 50 GB or 1,000 files. schemaLocation can share the checkpoint folder.

2. Default inference: strings

```
CSV, JSON, XML    every column inferred as STRING (plus _rescued_data)
Parquet, Avro     types from the file schema (widest type wins on conflict)
cloudFiles.inferColumnTypes = true   -> typed inference (int, double, ...)
```

Strings by default so a changing value can't break a column type later.

3. Schema hints pin known types

```
.option("cloudFiles.inferColumnTypes", "true")
.option("cloudFiles.schemaHints", "amount DECIMAL(10,2), order_date DATE")
```

Hints apply on top of inference, can add a column not seen yet, and are used only without .schema(...).

4. Write side: mergeSchema

```
  .writeStream
  .option("checkpointLocation", chk)
  .option("mergeSchema", "true")      # Delta table accepts new columns
  .trigger(availableNow=True)          # serverless: process files, then stop
  .toTable("orders_bronze")
```

Without mergeSchema the restart fails at the Delta write with a schema mismatch.

Evolution modes, _rescued_data, exam traps

5. cloudFiles.schemaEvolutionMode

```
addNewColumns (default)        fail once (UnknownFieldException), schema updated, restart resumes
addNewColumnsWithTypeWidening  same + widen types like int -> long (Public Preview)
rescue                         never evolve, never fail, new columns go to _rescued_data
failOnNewColumns               fail, stay failed until you update the schema or remove the file
none                           ignore new columns, never fail, no rescue unless rescuedDataColumn
```

Provided .schema(...): default mode is none and addNewColumns is not allowed.

6. Day 1 -> day 2 (new column channel + amount N/A)

```
run 1  orders_day1.csv              -> 3 rows, typed
run 2  orders_day2.csv + channel    -> stream stops, schema location gains channel, still 3
run 3  same code (restart)          -> 5 rows, channel null for day 1 rows
       order 1005 amount N/A        -> amount null, _rescued_data {"amount":"N/A","_file_path":...}
```

7. What lands in _rescued_data

Type mismatches, columns missing from the schema, and case mismatches, stored as a JSON blob with the source _file_path. With a rescued data column, type mismatches don't fail the read; only malformed records do. Rename it or add it to a provided schema with the rescuedDataColumn option.

8. Five exam traps

1) CSV and JSON infer strings unless cloudFiles.inferColumnTypes. 2) Default mode: a new column fails the stream once, a restart picks it up (run it in Lakeflow Jobs with retries). 3) rescue: no failure, no new column. 4) Provided schema: default mode is none. 5) New column in the Delta table: mergeSchema on the write.

Practice challenge

Day 3 adds a coupon column. Default mode: fails once, then the rerun loads day 3 and mergeSchema adds coupon. rescue: no failure, no column, coupon in _rescued_data. Provided schema, no mode: none, coupon ignored, not rescued unless rescuedDataColumn is set.

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